CLASS 11 MATHEMATICS
CHAPTER 1 : SETS
Complete Notes

CHAPTER ROADMAP

LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Understand the meaning of a set.
- Represent sets in different forms.
- Identify different types of sets.
- Represent sets using Venn diagrams.
- Perform operations on sets.
- Use important laws of sets.
- Understand Cartesian products.
- Understand the basic concept of relations.

1. BASIC CONCEPTS OF SETS

Definition: Set

A **set** is a well-defined collection of distinct objects.
The objects belonging to a set are called its **elements** or **members**.

Example

The collection of vowels in the English alphabet is
$A = \{a, e, i, o, u\}$.
Here a, e, i, o, u are the elements of the set A.

Well-Defined Collection

A collection must be well-defined to be called a set.
For example:

- The collection of first five natural numbers is a set.
- The collection of "beautiful flowers" is not well-defined, because the meaning of beautiful may differ from person to person.

Set Notation

Sets are generally represented by capital letters:
A, B, C, X, Y, Z.
Elements are generally represented by small letters:
a, b, c, x, y, z.
If x is an element of A, we write
$x \in A$.
If x is not an element of A, we write
$x \notin A$.

Remember

$x \in A$ means "x belongs to A".

$x \notin A$ means "x does not belong to A".

Representation of a Set

A set can be represented mainly in two ways:

1. Roster or Tabular Form
2. Set-Builder Form

1. Roster Form

In roster form, all the elements of a set are listed inside curly brackets.
For example,
$A = \{2, 4, 6, 8, 10\}$.
Another example is
$V = \{a, e, i, o, u\}$.

2. Set-Builder Form

In set-builder form, a set is described by stating the common property of its elements.
For example,
$A = \{x : x \text{ is an even natural number less than } 12\}$.
Therefore,
$A = \{2, 4, 6, 8, 10\}$.

Number of Elements of a Set

The number of distinct elements in a set A is called its **cardinality**.
It is denoted by $n(A)$ or $|A|$.
For example,
$A = \{2, 4, 6, 8, 10\}$.
Therefore,
$\boxed{n(A) = 5}$.

Quick Example

If
$A = \{1, 2, 3, 3, 4, 4, 5\}$,
then repeated elements are counted only once.
Hence,
$A = \{1, 2, 3, 4, 5\}$.
Therefore,
$\boxed{n(A) = 5}$.

2. TYPES OF SETS

1. Empty Set

A set containing no element is called an **empty set** or **null set**.
It is denoted by
$\emptyset$ or $\{\}$.
Example:
$A = \{x \in \mathbb{N} : x < 1\}$.
Hence,
$A = \emptyset$.

Important:
$\boxed{\emptyset \neq \{\emptyset\}}$
because
$n(\emptyset) = 0$
whereas
$n(\{\emptyset\}) = 1$.

2. Singleton Set

A set containing exactly one element is called a **singleton set**.
Example:
$A = \{5\}$.
Therefore,
$n(A) = 1$.

3. Finite Set

A set containing a finite number of elements is called a **finite set**.
Example:

$A = \{1, 2, 3, 4, 5\}$.

4. Infinite Set

A set having infinitely many elements is called an **infinite set**.
Example:
$\mathbb{N} = \{1, 2, 3, 4, \ldots\}$.

5. Equal Sets

Two sets A and B are equal if they contain exactly the same elements.
We write
$A = B$.
Example:
$A = \{1, 2, 3\}$
and
$B = \{3, 1, 2\}$.
Therefore,
$\boxed{A = B}$.

> **Remember:**
> The order of elements does not matter in a set.
> $\{1, 2, 3\} = \{3, 2, 1\}$.

6. Equivalent Sets

Two sets are called **equivalent sets** if they have the same number of elements.
If
$n(A) = n(B)$,
then A and B are equivalent sets.
Example:
$A = \{1, 2, 3\}$
and
$B = \{a, b, c\}$.
Thus,
$n(A) = n(B) = 3$.

7. Subset

A set A is called a subset of B if every element of A is also an element of B.
It is written as
$A \subseteq B$.
Example:
$A = \{1, 2\}$
and
$B = \{1, 2, 3, 4\}$.
Therefore,
$\boxed{A \subseteq B}$.

8. Proper Subset

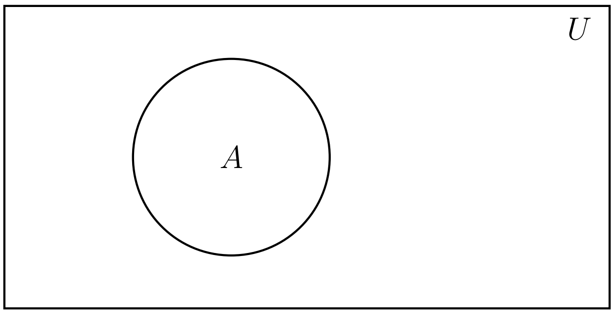
If
$A \subseteq B$
and
$A \neq B$,
then A is called a **proper subset** of B.

9. Universal Set

A set containing all the objects under consideration is called the **universal set**.
It is usually denoted by U.

3. VENN DIAGRAMS

A **Venn diagram** is a pictorial representation of sets.
Usually, the universal set is represented by a rectangle and a set is represented by a closed curve.



4. OPERATIONS ON SETS

The important operations on sets are:

1. Union
2. Intersection
3. Difference
4. Complement

Union of Two Sets

The union of sets A and B is the set of all elements which belong to A or B or both.
It is denoted by
$A \cup B$.
Thus,
$A \cup B = \{x : x \in A \text{ or } x \in B\}$.

Example:
Let
$A = \{1, 2, 3\}$
and
$B = \{3, 4, 5\}$.
Then,
$A \cup B = \{1, 2, 3, 4, 5\}$.

Intersection of Two Sets

The intersection of A and B is the set of elements common to both sets.
It is denoted by
$A \cap B$.
Thus,
$A \cap B = \{x : x \in A \text{ and } x \in B\}$.
For the above example,
$A \cap B = \{3\}$.

Difference of Two Sets

The difference of A and B is the set of elements which belong to A but do not belong to B.
It is denoted by
$A - B$.
Thus,
$A - B = \{x : x \in A \text{ and } x \notin B\}$.
For
$A = \{1, 2, 3\}$
and
$B = \{3, 4, 5\}$,
we get
$A - B = \{1, 2\}$.
Similarly,
$B - A = \{4, 5\}$.

Complement of a Set

If U is the universal set and $A \subseteq U$, then the complement of A is
$A' = U - A$.
It contains all elements of U which are not in A.

KEY FORMULA

$\boxed{A' = U - A}$

5. LAWS OF ALGEBRA OF SETS

Law	Result
Commutative	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A$ $A \cap U = A$
Domination	$A \cup U = U$ $A \cap \emptyset = \emptyset$
Idempotent	$A \cup A = A$ $A \cap A = A$
Complement	$A \cup A' = U$ $A \cap A' = \emptyset$
Double Complement	$(A')' = A$
De Morgan	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

6. CARTESIAN PRODUCT

Let A and B be two non-empty sets.
The Cartesian product $A \times B$ is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$.
Thus,
$A \times B = \{(a, b) : a \in A,\ b \in B\}$.

Example

Let
$A = \{1, 2\}$
and
$B = \{3, 4\}$.
Then,
$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$.
Therefore,
$n(A \times B) = 4$.
Also,
$n(A) = 2, \qquad n(B) = 2$.
Hence,

$$\boxed{n(A \times B) = n(A) \times n(B)}$$

In general,
$A \times B \neq B \times A$.

7. RELATIONS

Let A and B be two sets.
A **relation** from A to B is any subset of $A \times B$.
Thus,
$R \subseteq A \times B$.

Example

Let
$A = \{1, 2\}$
and
$B = \{3, 4\}$.
Then,
$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$.
Consider
$R = \{(1, 3), (2, 4)\}$.
Since
$R \subseteq A \times B$,
R is a relation from A to B.

IMPORTANT QUESTIONS

1. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, find $A \cup B$, $A \cap B$, $A - B$ and $B - A$.
2. Show that $A \cap B = A \cap C$ need not imply $B = C$.
3. Prove that $(A \cup B)' = A' \cap B'$.
4. Prove that $(A \cap B)' = A' \cup B'$.
5. Let A, B and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$. Show that $B = C$.

QUICK REVISION BOX

- $x \in A$ means x belongs to A.
- $A \subseteq B$ means every element of A belongs to B.
- $A \cup B$ = elements in A or B.
- $A \cap B$ = common elements of A and B.
- $A - B$ = elements in A but not in B.
- $A' = U - A$.
- $n(A \times B) = n(A)n(B)$.
- $R \subseteq A \times B$ represents a relation from A to B.

Understand the Set, and Mathematics becomes easier!

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